



Estimating Player Impact in College Basketball via Regularized Adjusted Plus-Minus

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Background

- Regularized Adjusted Plus-Minus (RAPM) is one of the most widely used metrics in basketball analytics.
- RAPM measures player impact on a team's point differential per 100 possessions. Unlike standard box-score stats, RAPM accounts for the quality of teammates and opponents on the floor.
- RAPM is readily available for the NBA, but not as much for college basketball. Reasons for this gap include the much larger number of players and the messy, less available play-by-play data.

Methods

- Transformed raw play-by-play data into $X \in \{0,1\}^{N \times 2M}$, where N is the number of possessions, and M is the number of players in our data (offensive and defensive coefficient for each player). 0 means a player is off the floor and 1 means they were on the floor for that possession.
- $y_i \in \{0,1,2,3, \dots\}$: points score on possession i .
- From here, we apply a ridge regression model to solve for the coefficients: $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$
- We multiply by 100 to obtain an estimate for a player's impact per 100 possessions. To make the numbers more intuitive, defensive RAPM is negated so a positive number is better.
- Total RAPM = ORAPM + DRAPM

Why Ridge Regression?

- Low-minute players introduce significant noise, and pushing these coefficients towards zero helps to reduce this.
- Since lineups are often repeated for many possessions, there is significant multicollinearity between players on the same team.

Improving the Raw Estimate

- Instead of pushing all coefficients to 0, we can push them to a prior "guess" of each player's true impact to make estimates more stable and converge faster.
- Here, offensive and defensive BPM values were used as prior estimates, scaled by a value dependent on a player's playing time.

RAPM Flavors

- Raw RAPM:** RAPM without the prior estimate. Useful for identifying high-minute players whose impact goes beyond the box score.
- Bayesian RAPM:** RAPM with the prior estimate. Useful for obtaining a more comprehensive ranking of the best players in college basketball.
- Multi-Season RAPM:** RAPM calculated over the course of multiple seasons. Useful for discussing the overall careers of veteran players.

Data was scraped from College Basketball Reference and stats.ncaa.org. Portal data provided by collegebasketballdata.com and 247 Sports.

Potential Use Cases

Finding Potential Breakout Stars

Jaden Bradley started 0 games for Arizona in 2023-24...

Rk	Player	Pos	G	GS	MP	FG	FGA	FG%
1	Caleb Love	G	36	36	32.3	6.1	14.7	.413
2	Pelle Larsson	G	36	36	30.1	4.5	8.7	.519
3	Keshad Johnson	F	36	36	27.6	4.2	7.9	.530
4	Kylan Boswell	G	36	35	26.9	3.5	8.8	.396
5	Oumar Ballo	C	36	36	26.0	5.1	7.8	.658
6	Jaden Bradley	G	36	0	20.3	2.4	5.2	.462
7	KJ Lewis	G	36	0	18.3	2.1	4.5	.466
8	Motiejus Krivas	C	36	0	12.1	2.0	3.6	.554

...but was Arizona's best player according to raw RAPM

Player	ORAPM	DRAPM	Total RAPM
Jaden Bradley	3.4599	4.497	7.9569
Pelle Larsson	4.3242	2.3123	6.6365
Caleb Love	2.0293	3.1054	5.1347
Motiejus Krivas	2.1111	2.68	4.7911
Oumar Ballo	2.433	2.1061	4.5391
Keshad Johnson	3.0063	1.4715	4.4779
KJ Lewis	2.6046	1.1055	3.7101
Kylan Boswell	2.6546	0.4882	3.1428
Filip Borovicarin	1.2213	1.3562	2.5775
Conrad Martinez	1.1497	0.4604	1.6101
Grant Weitman	0.4353	0.2377	0.673
Paulius Murauskas	1.0144	-0.3604	0.654
Luke Champion	0.3404	0.0663	0.4066
Will Menaugh	0.4201	-0.3769	0.0432

Two seasons later:

TUCSON, Ariz. – The Big 12 Conference announced the end of year awards for the 2025-26 season on Monday, as voted on by the coaches in the league, and Arizona senior point guard **Jaden Bradley** was named the Player of the Year after guiding the Wildcats to a 29-2 overall record and the Big 12 regular season championship.

Limitations/Caveats

- RAPM takes a very long time to stabilize, and many college players will never play enough minutes to obtain ideal estimates.
- Single-season RAPM numbers can be extremely noisy, especially for players on teams with injured stars.
- Difficult to use Multi-Season RAPM to compare four-year players with one-and-dones due to the massive difference in possessions played.

Identifying Value in the Transfer Portal

Correctly Identified Underrated 2025-26 Transfers					
Player	Transferred From	Transferred To	247 Transfer Portal Rank	Bayesian RAPM Rank	Rank Difference
Henri Veessaar	Arizona	North Carolina	36	1	+35
Pryce Sandfort	Iowa	Nebraska	187	23	+164
Bennett Stürtz	Drake	Iowa	12	3	+9
Dillon Mitchell	Cincinnati	St. John's	42	10	+32
Donovan Atwell	UNC Greensboro	Texas Tech	170	98	+72
Ja'Kobi Gillespie	Maryland	Tennessee	16	4	+12
Duke Miles	Oklahoma	Vanderbilt	202	78	+124
Bryce Lindsay	James Madison	Villanova	277	174	+103
Aday Mara	UCLA	Michigan	49	27	+22

Correctly Identified Overrated 2025-26 Transfers					
Player	Transferred From	Transferred To	247 Transfer Portal Rank	Bayesian RAPM Rank	Rank Difference
Ian Jackson	North Carolina	St. John's	7	639	-632
Joseon Sanon	Arizona St.	St. John's	38	284	-246
Simeon Wilcher	St. John's	Texas	94	464	-370
AJ Storr	Kansas	Mississippi	47	1041	-994
Pop Isaacs	Creighton	Texas A&M	39	256	-217
Jalil Bethea	Miami FL	Alabama	52	1251	-1,199
Kennard Davis	Southern Illinois	BYU	41	836	-795
Wesley Yates III	USC	Washington	33	112	-79

Top 2026-27 Players in the Portal as of April 15th, 2026						
Player	Position	Origin School	Destination School	Multi-Year Bayesian RAPM	2026 Bayesian RAPM	247 Transfer Portal Rank
Flory Bidunga	C	Kansas	Louisville	13.77	14.05	1
Milan Momcilovic	PF	Iowa St.		11.61	12.13	5
John Blackwell	SG	Wisconsin		11.09	7.88	2
Mouhamed Dioubate	SF	Kentucky		11.02	7.19	70
Mikey Lewis	G	Saint Mary's		10.27	6.28	119
Tyrone Riley	SF	San Francisco	Oregon	10.23	8.11	134
PJ Haggerty	PG	Kansas St.	Texas A&M	10.06	4.48	9
Miles Byrd	SF	San Diego St.	Providence	9.92	7.23	13
Trevian Carson	G	North Dakota St.		9.69	8.64	338
Terrance Arceneaux	SF	North Carolina St.	George Washington	9.64	2.50	7
Aiden Sherrell	PF	Alabama	Indiana	9.54	8.91	29
Emmanuel Innocenti	SF	Gonzaga	Arizona State	9.27	6.96	302
Jasper Floyd	SG	West Virginia		9.19	6.49	Out of Eligibility
Allen Graves	PF	Santa Clara		9.06	8.98	4
Jackson Holcombe	SG	Utah Valley	Utah	9.00	8.51	227

Future Directions

- Expand historical database
- Use separate penalization values for offensive and defensive RAPM estimates
- Analyze overarching transfer portal trends, such as if RAPM is a useful predictor of which players succeed when transferring from mid or low-major schools to power conferences

Full Database/Resources

